

Business Process Technologies and Management

Exercises

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Assignment 6 - What to do (total 20 points)?

1. Export BPMN 2.0 XML from Signavio
2. Create HTTP Message Correlation Service
 - Create a correlator service, that
 - stores messages upon POST (Send Message)
 - Parameter 1: message ID
 - Parameter 2: storage policy
 - Parameter 3: payload
 - delivers messages upon GET (Request Message Delivery)
 - Parameter 1: message ID
 - Parameter 2: retrieval policy
 - The service works strictly asynchronous for GET and POST (CPEE-CALLBACK: true)
 - Data Retention and policies:
 - When a message with a certain message id is requested (GET) and it is already stored (compare message id), deliver the message
 - When a message with a certain message id is requested (GET) and it is NOT stored (compare message id), wait
 - When a message is sent and no requests for the message id exist, store the message
 - When a message is sent and requests for the message id exist, deliver the message
 - Storage policy “replace”: when a message id already exists, replace messages
 - Storage policy “queue”: when a message id already exists, add the message to a queue
 - Retrieval policy “multi instance”: messages have to be stored indefinitely, so that if different instances ask for the value, they can be delivered in order of arrival.
 - Retrieval policy “multi task”: messages have to be stored indefinitely, so that if different tasks ask for the value, they can be delivered in order of arrival.
 - Retrieval policy “once”: message is delivered and then immediately deleted.
 - If different retrieval policy requests for the same message id once > multi task > multi instance
3. Please test it:
 - Inter-instance communication - add a way to send message to between either (1) processes and subprocesses, or (2) between instances of the same process
 - Intra-instance communication - use the method to synchronize parallel branches

Assignment 6

Submission Artefacts:

1. Export BPMN 2.0 XML from Signavio - newest version
2. Testset with Sample Process
3. Source Code (ZIP)

Deadline: deadlines for assignment 5 and 6 are 15 days after the end of the semester.

Please have first results in 1 week, so that we can discuss and answer questions in the course.